

Development and Preliminary Evaluation of a Locally Deployed Python-based Retrieval-Augmented Language AI Model for Educational and EMS Decision Support

Author: Dominique Ball

ORCID: 0009-0007-6689-6363

Corresponding Author Email: Domhasaball@gmail.com

Article Type: Original Research Article

Abstract

Emergency medical services (EMS) associates regularly make time-sensitive clinical decisions applying knowledge obtained from patient history, symptoms, vital signs, electrocardiographic findings, medications, and environmental or situational factors. With recent advancements in large language models (LLMs), artificial intelligence (AI)-based decision-support systems gain the opportunity to be utilized. Recently, concerns regarding hallucination, inaccurate clinical solutions, outdated information and education, and lack of source traceability limit their direct application to medical environments.

Objective

This study aims to advance and preliminarily evaluate a locally deployed, retrieval-augmented artificial intelligence system engineered to provide educational and decision-support responses to simulated paramedic scenarios.

Methods

A prototype application was developed using Python and a locally hosted large language model using Ollama. The system accepts clinical scenarios containing symptoms, vital signs, electrocardiographic information, patient's characteristics (age, height, weight, sex, and age), medication-related questions, and situational descriptions. A retrieval-augmented generation (RAG) system searches a deeply curated knowledge base for relevant EMS reference material. The retrieved documentations are supplied to the language model as contextual information before response generation begins. The artificial intelligence system additionally provides retrieved references with generated responses and instructs the model not to fabricate protocols or medication doses and to defer to current local EMS protocols and medical direction. The AI LLM prototype has been evaluated using standardized datasets of simulated EMS scenarios. The AI-generated feedback will be assessed for clinical correctness, protocol consistency, medication accuracy, ECG interpretation, source consistency, recognition of potentially life-threatening conditions, completeness, and unsafe recommendations.

Results

Results were recorded and assessed following the completion of the experimental evaluation. Planned measured assessments include overall clinical performance, medication accuracy, protocol compliance, documentation retrieval performance, and frequency of unsafe or unsupported conclusions and solutions.

Conclusion

This study will investigate the practical application of using a locally deployed retrieval-augmented AI LLM as an educational and EMS decision-support prototype. The study is to deliberately characterize system performance, identify failure, and analyze whether retrieval of

relevant EMS reference material improves response quality. The prototype has no intended place to replace paramedics, physicians, medical direction, or established EMS protocols.

Keywords: artificial intelligence; large language models; emergency medical services; paramedicine; retrieval-augmented generation; clinical decision support; machine learning; EMS; medical AI

1.1 Background

Emergency medical services demand clinicians exercise rapid decision-making under conditions of ambiguity and limited information. Paramedics may be obligated to analyze patient symptoms, vital signs, electrocardiographic findings, medication histories, cardiac murmurs, and environmental or situational elements while simultaneously determining appropriate assessment and treatment interventions.

Artificial intelligence and retrieval-augmented generation have increasingly been evaluated as a method for supporting clinical decision-making. Large language models are competent at processing natural-language information and generating responses to complex situational inquiries. However, general-purpose language models may generate inaccurate or unsupported documented information, notably in specialized clinical environments.

One potential methodology to reduce the limitations is retrieval-augmented generation. RAG systems extract relevant information from an external knowledge base and provide that

contextual information to a language model for pre-generation of an evidence-based response. This framework may improve source grounding and reduce reliance on information contained solely within the model's parameters.

1.2 Research Problem

Although LLMs demonstrate ample language and reasoning capabilities, their use in EMS decision support illustrates several costly considerations. These include inaccurate clinical reasoning, medication errors, hallucinated protocols, failure to recognize uncertainty, and the possibility of providing recommendations that conflict with local EMS procedures.

Clinical correctness, protocol consistency, source traceability, and potentially harmful recommendations must be examined independently.

1.3 Study Objective

The primary objective of this study is to establish and preliminarily analyze and research a locally deployed retrieval-augmented AI system coherent with responding to simulated paramedic scenarios.

Secondary objectives are to:

1. Evaluate the clinical correctness of generated responses.
2. Evaluate medication-related recommendations.
3. Evaluate responses to vital-sign and symptom-based scenarios.
4. Evaluate responses to ECG-related questions.
5. Determine whether retrieved reference material improves response quality.

6. Evaluate the frequency and characteristics of unsafe recommendations.
7. Evaluate the ability of the system to identify uncertainty and defer to appropriate clinical authority.
8. Evaluate the accuracy of retrieved reference material.

1.4 Hypothesis

Primary Hypothesis

The retrieval-augmented configuration will demonstrate higher clinical and protocol-consistency scores than the non-retrieval configuration

Null Hypothesis

There will be no statistically significant difference in clinical and protocol-consistency assessments between the retrieval-augmented system and the non-retrieval language-model condition.

2. Materials and Methods

2.1 System Design

The prototype was engineered using the Python programming language. A graphical user interface was developed using the Tkinter model. The system application provides an interface with which users can enter clinical questions and simulate in-field patient scenarios. \

The application system was developed to process:

- Patient symptoms
- Vital signs
- ECG-related information
- Medication-related questions
- Clinical scenarios
- Situational EMS questions
- General paramedic decision-support questions

The overall system consists of four primary components:

1. User interface
2. Knowledge retrieval system
3. Locally hosted language model
4. Session and response management

The application system documentation contains:

Area	Approx. page references
EMS clinical decision-making	3–5
LLMs in medicine	4–6
LLM hallucination/reliability	3–5
RAG methodology	3–5

RAG in healthcare/clinical settings	3–6
AI clinical decision support	3–5
ECG/clinical evaluation methodology	2–4
Signs and Symptoms Assessment	37
Reference	
AI reporting guidelines	35
Clinical thinking tree	26
Airway Assessment and Initial Management	47
Cardiology Assessment and Initial Management	21
Prehospital Medication Management	22
Pediatric Assessment and Initial Management	30
Protocol Management and Clinical Decision Reference	25
EMS Training and Education	30
Reference	

EMS Trauma — Clinical Reference	31
& Decision Support	
ECG / EKG Interpretation Reference	42
EMS Reference	70
Prehospital Medications and Oxygen	35
Reference	
PARAMEDIC MEDICATION	51
CLINICAL DECISION SUPPORT	
PARAMEDIC TRIAGE	54
REFERENCE DOCUMENT	
Total	592

2.2 Large Language Model

The prototype uses a locally hosted language model through the Ollama application. The model is accessed programmatically through the Python Ollama interface. Local model implementation was designated to allow the prototype to operate without requiring transmission of clinical scenario information to an external cloud-based language model service. The exact model identifier, model version, hardware configuration, and inference parameters will be recorded to facilitate reproducibility.

2.3 Retrieval-Augmented Generation

The programmed system integrates a retrieval-augmented generation architecture. When a user submits a question, the question is passed to the knowledge-retrieval component. Relevant documents are returned and implemented into the context allocated to the language model.

Retrieved information uses metadata, including:

- Document title
- Source
- Jurisdiction
- Document type
- Effective date
- Status
- Retrieved text

The retrieved resources are presented to the language model as reference information. The system instructs the model to use relevant retrieved findings, avoid fabrication of protocol information and medication doses, and recognize that current local EMS protocols and medical direction take precedence during patient care.

2.4 Knowledge Base

The knowledge base consists of EMS-related reference material used to contribute contextual information to the language model. Each document is correlated with metadata describing its

source, jurisdiction, document type, effective date, and status. The knowledge base is reviewed before experimental evaluation to identify duplicate, outdated, or incompatible material.

2.5 Experimental Dataset

A standardized dataset of simulated EMS scenarios was produced for evaluation.

Scenarios will represent multiple clinical categories, including:

- Cardiovascular emergencies
- Respiratory emergencies
- Neurological emergencies
- Trauma
- Medication-related scenarios
- ECG interpretation
- Shock and hemodynamic instability
- Pediatric emergencies
- Environmental emergencies
- Complex or mixed presentations

Each scenario will contain standardized patient information and one or more questions requiring the AI system to generate a response.

2.6 Reference Standards

Each scenario will have a reference standard from appropriate authoritative EMS or medical sources. The reference standard will identify clinically important aspects expected in an

appropriate feedback, including relevant assessment considerations, treatment considerations, medication requirements where applicable, contraindications, and appropriate escalation or medical-direction considerations. Reference standards will be established before evaluation of the AI system.

2.7 Experimental Conditions

Two experimental conditions will be evaluated.

Condition A: Language Model Without Retrieval

The scenario will be provided directly to the language model without retrieved reference material.

Condition B: Retrieval-Augmented Paramedic AI

The same scenario will be processed through the retrieval system. Relevant reference material will then be supplied to the language model before the response generation system. Both conditions will receive identical scenario information. This experimental design will allow evaluation of whether retrieval augmentation improves system performance relative to the non-retrieval condition

2.8 Evaluation Criteria

Responses will be evaluated using a previously determined scoring rubric.

Primary evaluation categories will include:

1. Clinical correctness

2. Completeness
3. Protocol consistency
4. Medication accuracy
5. ECG interpretation
6. Recognition of life-threatening conditions
7. Appropriate handling of uncertainty
8. Reference/source consistency

Each response will additionally be evaluated for potentially unsafe recommendations. Clinical correctness (100 questions) Completeness (100 questions) Protocol consistency (100 questions) Medication accuracy (50 questions) ECG interpretation (25 questions) Recognition of life-threatening conditions (10 questions) Appropriate handling of uncertainty (100 questions) Reference/source consistency (100 questions).

2.9 Safety Evaluation

A separate safety analysis will be determined to identify high-risk errors.

Potential error categories will include:

- Incorrect medication
- Incorrect medication dose
- Contraindicated intervention
- Failure to recognize a life-threatening condition
- Incorrect ECG interpretation
- Fabricated protocol information

- Unsupported clinical claims
- Failure to acknowledge insufficient information
- Inappropriate confidence
- Incorrect interpretation of retrieved material

The unsafe-response rate will be calculated as the number of responses containing a predefined unsafe recommendation divided by the total number of evaluated responses.

2.10 Retrieval Evaluation

The retrieval component will be evaluated independently of the language model. For each scenario, evaluators will determine whether an appropriate citation was retrieved amid the highest-ranked results. Retrieval performance may be reported using measures such as recall at K (Recall@K), where K represents the number of retrieved documents evaluated.

2.11 Statistical Analysis

Descriptive statistics will be calculated for all evaluation results. Based on the final study design and data distribution, comparative statistical tests will be selected to evaluate differences between the retrieval and non-retrieval conditions. Continuous variables may be summarized using means and standard deviations or medians and interquartile ranges, as appropriate. Categorical outcomes will be reported as frequencies and percentages.

Statistical significance will be assessed using a predefined significance threshold.

2.12 Reproducibility

The following technical characteristics will be documented:

- Python version
 - Operating system
 - Hardware configuration
 - Ollama version
 - Language-model version
 - Retrieval implementation
 - Knowledge-base contents
 - Prompt configuration
 - Model inference parameters
 - Number and type of evaluation scenarios
 - Evaluation rubric
 - Statistical methodology
-

3. Results

3.1 Overall Performance

The prototype was evaluated using 100 standardized test scenarios. The retrieval-augmented generation (RAG) configuration produced correct responses in 92 of 100 cases, corresponding to an overall accuracy of **92%**. In comparison, the language model operating without retrieval produced correct responses in 71 of 100 cases, corresponding to an overall accuracy of **71%**.

The retrieval-augmented configuration therefore demonstrated a **21-percentage-point improvement** in overall accuracy compared with the non-RAG configuration. In relative terms, the RAG configuration produced approximately **29.6% more correct responses** than the non-RAG configuration.

Evaluation condition	Correct responses	Accuracy
RAG	92/100	92%
No RAG	71/100	71%
Difference	21 cases	+21 percentage points

3.2 Medication Performance

The system achieved **100% accuracy** on the medication-related test cases included in the evaluation dataset.

No incorrect medication recommendations were identified within the evaluated medication cases. This result indicates that the system successfully generated responses consistent with the reference standards used for the medication evaluation.

Because medication errors can have significant clinical consequences, the absence of medication errors in this test set represents a positive finding. However, the result should be interpreted within the limits of the test dataset and should not be considered evidence of universal medication safety or clinical readiness. In all results the AI system reminds the user to refer to the local protocol and the local medical director.

3.3 ECG Performance

The system achieved an ECG-related accuracy of **91%** on the evaluated ECG scenarios.

This result indicates that the prototype correctly addressed the large majority of ECG-related test cases. However, approximately 9% of ECG cases were not scored as fully correct, demonstrating that ECG-related reasoning remains an area requiring further investigation and error analysis.

3.4 Safety Performance

No unsafe responses were identified during the evaluation. The system produced **0 identified unsafe responses across the 100 evaluated scenarios**, corresponding to an observed unsafe-response rate of **0%**. The prototype started every directional simulation with the standard BSI PPE scene safety procedure.

Although this finding is encouraging, a zero observed unsafe-response rate does not establish that the system is inherently safe for clinical deployment. The result reflects performance within the specific scenarios and evaluation criteria used in this study. Additional testing using larger and more challenging datasets would be required to determine the robustness of this finding.

3.5 Error Analysis

The RAG configuration produced 8 incorrect responses out of the 100 evaluated scenarios, while the non-RAG configuration produced 29 incorrect responses.

The difference in error frequency was therefore substantial. The RAG configuration reduced the number of incorrect responses by **21 cases**, representing a reduction of approximately **72.4% in observed errors** compared with the non-RAG configuration.

Further analysis of the eight incorrect RAG responses should be conducted to determine whether the errors resulted from language-model reasoning, incomplete information, retrieval failure, interpretation of retrieved material, or other factors.

3.6 Summary of Findings

The preliminary evaluation exhibited a higher overall performance when the language model was supplemented with retrieved referenced material. The RAG configuration achieved 92% accuracy compared with 71% without retrieval. Medication-related scenarios achieved 100% accuracy, while ECG-related scenarios achieved 91% accuracy. No unsafe responses were identified in the evaluated test set.

Overall, these findings support further evaluation and analysis of retrieval-augmented language models for paramedic-oriented educational and decision-support application systems. However, the results reflect a prototype evaluation and do not establish clinical efficacy, clinical safety, or readiness for independent use in patient care.

4. Discussion

4.1 Principal Findings

The primary finding of this study was that the retrieval-augmented generation (RAG) configuration substantially outperformed the non-RAG configuration in the evaluated paramedic scenarios. The RAG system achieved an overall accuracy of 92%, compared with 71% for the language model without retrieved reference material. This represents a 21-percentage-point difference in accuracy.

These findings suggest that providing the language model with relevant external reference material may improve its ability to generate responses that are consistent with the information contained within the knowledge base. The improvement observed in this prototype supports the hypothesis that retrieval augmentation can contribute to improved performance in paramedic-oriented decision-support scenarios.

4.2 Medication-Related Performance

The system achieved 100% accuracy on the medication-related scenarios included in the evaluation. This was an important finding because medication recommendations represent a potentially high-risk component of clinical decision support.

The absence of inconsistent medication responses in the measured dataset suggests that the prototype was able to provide responses that accurately reflected the standards used in the study. The results are not to be interpreted as evidence that the system will always provide correct medication information.

Medication recommendations can depend on patient factors such as patient age, weight, allergies, contraindications, medical history, local protocols, and available medications. Future assessments, therefore, should include a larger variety of medication scenarios, including contraindications, incomplete patient information, pediatric cases, and situations involving conflicting or outdated information.

4.3 ECG Performance

The prototype achieved 91% accuracy on ECG-related scenarios. This assessment of relatively strong performance identified ECG interpretation as an area requiring additional investigation. The 9% of ECG cases that were not fully correct are particularly important because fault in ECG interpretation will eventually have significant consequences in emergency medical decision-making. Future research should therefore examine these inconsistencies individually to identify whether they originated from incorrect interpretation, insufficient contextual information, retrieval limitations, or language-model reasoning. Additional evaluation using standardized ECG datasets and expert-reviewed interpretations would provide a stronger assessment of the system's ECG capabilities.

4.4 Safety Considerations

No unsafe responses were identified among the 100 evaluated scenarios. This represents an observed unsafe-response rate of 0% within the study dataset.

Although this result is encouraging, it should be interpreted cautiously. A zero observed error rate does not demonstrate that the system is clinically safe. The number and design of test scenarios determine the types of errors that can be detected. Based on how the system was

engineered, the AI prompt always assesses the BSI scene safely in any scenario-based question. Rare but serious errors may not appear during a relatively small evaluation. Future testing should therefore include specifically designed safety and adversarial scenarios that try to push the RAG limitations. These could include incomplete patient information, contradictory vital signs, medication contraindications, unusual presentations, ambiguous questions, and scenarios in which the safest response is to acknowledge uncertainty or defer to medical direction.

4.5 Effect of Retrieval-Augmented Generation

The difference between the two experimental conditions provides preliminary evidence that retrieval augmentation improved the performance of the language model for paramedic-oriented questions.

Without retrieval, the model achieved 71% accuracy. With retrieval, accuracy increased to 92%. The reduction in incorrect responses from 29 to 8 suggests that access to external reference is one possible explanation: retrieval provides the model with information that is more directly relevant to the specific clinical question. This may reduce reliance on information encoded within the model and provide additional context for protocol-related questions.

However, retrieval augmentation does not exclude the possibility of inaccurate responses. The RAG system still produced eight unacceptable responses in the assessed dataset. This demonstrates that providing reference material alone is deficient in guaranteeing absolute clinical reasoning.

4.6 Clinical Implications

The findings suggest that a locally deployed AI system could potentially serve as an educational or decision-support tool for simulated paramedic scenarios. The local engineered architecture also provides advantages for research environments where control over the software, model, and knowledge base is desirable.

The AI system's ability to retrieve and display supporting referencing documents may also provide an advantage over an unrestricted conversational model because users can examine the sources used to generate an answer when referred to documentations come from the nationally accredited protocol. However, source retrieval does not guarantee that the generated interpretation of a source is correct.

Importantly, the prototype should be viewed as an **assistive system rather than an autonomous clinical decision-maker**. Current EMS protocols, qualified clinicians, and medical direction should remain the authoritative sources for actual patient care.

4.7 Future Research

Future research should expand the evaluated dataset and include a broader range of numbers of autonomously developed EMS scenarios. Additional research should also introspect the eight incorrect RAG responses in detail to identify recurring failure modes.

A larger study could evaluate:

1. Performance across additional EMS specialties.

2. More complex and ambiguous scenarios.
 3. Pediatric and weight-based medication scenarios.
 4. Contraindication and adverse-event scenarios.
 5. Additional ECG datasets.
 6. Retrieval accuracy and source ranking.
 7. Performance using different language models.
 8. Performance across different EMS jurisdictions.
 9. Adversarial and safety-focused testing.
 10. Inter-rater agreement among clinical evaluators.
-

5. Limitations

Several limitations are anticipated. First, the system represents a prototype rather than a clinically approved validated medical device. Second, simulated scenarios may not reproduce the complexity and uncertainty of real patient contacts. Third, language-model outputs may vary depending on model configuration and prompt conditions. The quality of the retrieval system is also dependent on the completeness, accuracy, and currency of its knowledge base. Personnel question adequacy also plays a part in the AI system's ability to regurgitate a precise result. Additionally, performance on a controlled research dataset may not generalize to real-world EMS environments. Finally, this study does not establish that the system can independently diagnose or treat patients and should not be interpreted as evidence that AI can replace paramedics, physicians, or medical direction.

6. Ethical and Safety Considerations

The initial evaluation will use simulated patient scenarios rather than identifiable patient information.

The system will be evaluated as an educational and research decision-support prototype rather than an autonomous clinical system.

During development and evaluation, particular emphasis will be placed on identifying potentially harmful recommendations and determining whether the system appropriately communicates uncertainty.

Any future evaluation involving real patient data or clinical deployment would require appropriate privacy protections, institutional oversight, and applicable regulatory and ethical review.

7. Conclusion

This study proposes the development and preliminary evaluation of a locally Python-based deployed retrieval-augmented artificial intelligence system for paramedic-oriented decision support. The prototype combines a Python-based interface, a knowledge-retrieval system, and a locally hosted large language model to generate responses to simulated EMS scenarios.

The proposed evaluation assessed clinical correctness, protocol consistency, medication-related performance, ECG interpretation, retrieval quality, response time, and safety-related failure modes. Comparison between retrieval-augmented and non-retrieval conditions determined that access to external EMS-referenced documentation from national protocol improved the quality and reliability of generated responses.

The primary purpose of the system is research and evaluation. Successful prototype performance justified that further investigation would be an advancement into the use of AI systems in the medical field but would not, by itself, establish clinical safety, efficacy, or suitability for autonomous patient care.